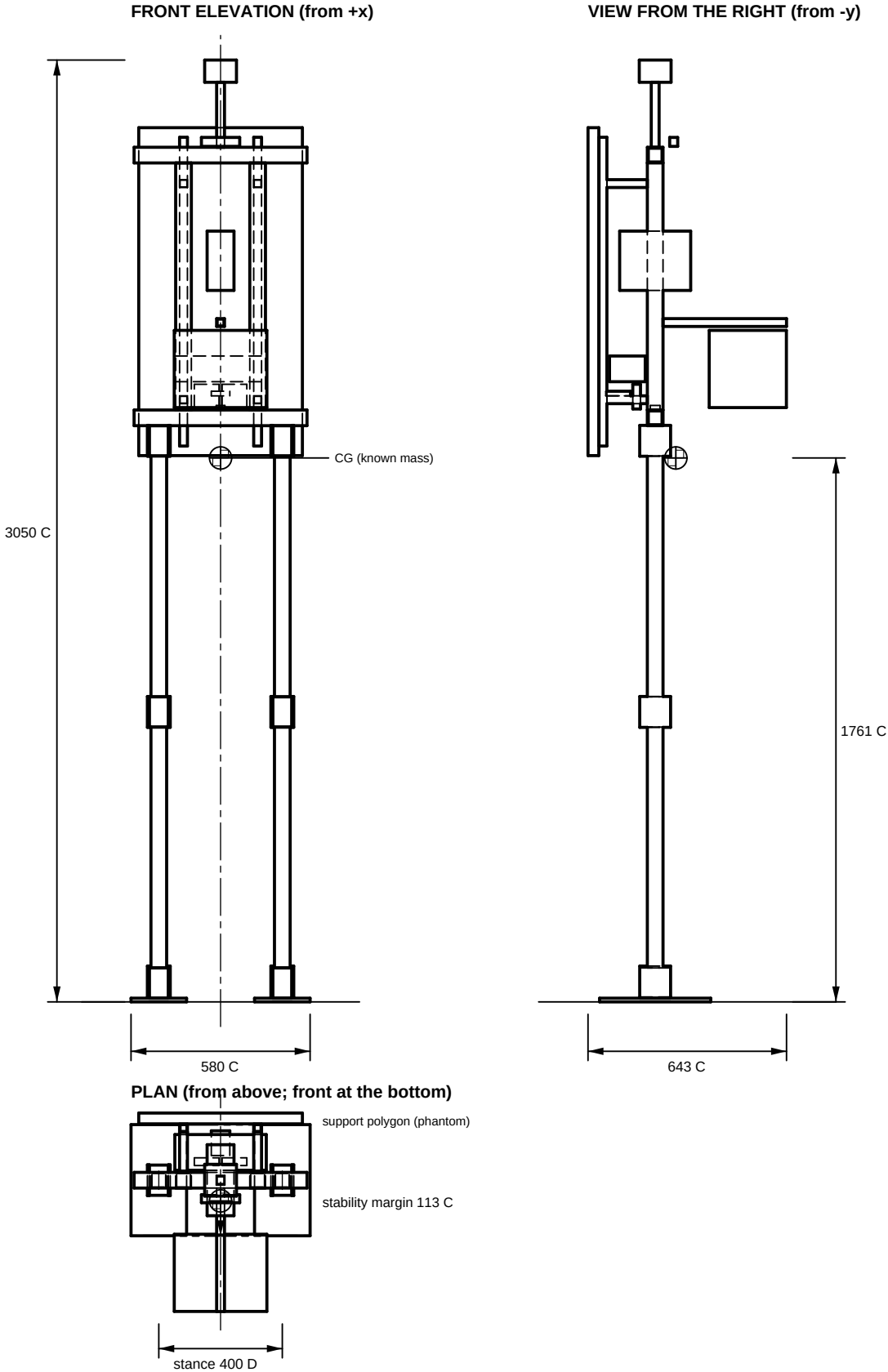




NOTES

Dimensions in mm. C = computed from the model, D = layout design choice, S = sourced (cited).
Values, bases and sources: sheet 2. Nothing is scaled off this sheet.
Bodies are drawn as the boxes the solvers use: frame tubes and plates at their section and length; bought parts (actuators, sensors, compute, battery) as datasheet envelopes, not detailed geometry. Hidden lines exact for those boxes. Joints, fasteners, wiring: not drawn.
Not drawn (no position or geometry in the model): armor, fasteners, wiring.
CG = centre of the KNOWN mass only. Unknown masses are listed on sheet 2; they can move it.
Revision = hash of the model this sheet shows. Any change to a part, position, mass or check gives a new revision; a sheet whose revision differs from the model's is superseded.
SCREENING DRAWING: generated from a software model; not checked, not approved, not for manufacture.
Masses: mass.budget states (sourced / computed / estimated / requirement / unknown / not fitted).
Passing software checks is not physical validation: weigh the parts and run the tip test.

GENERAL ARRANGEMENT - Sentinel M1 ground prototype		
Drawing no. CK-GA-SENTINEL-M1 Sheet 1 of 2	Revision	
Scale 1:20 (A3) Units mm First-angle projection	R-BD6B4261	
Model: sentinel: 46 part boxes + BOM (34 lines) + mass.budget@sentinel + stability.s...	model hash bd6b42619415a96d716a67d42b3d6062	
SCREENING - NOT FOR MANUFACTURE	ec93e07e11197396d7a5a40c414d1c41	
Generated by ConKay drawing.ga-assembly 1.0.0 from the design graph; revision = model hash		

BILL OF MATERIALS (part records; mass state per line)

Item	Qty	Nodes	Description	Material / part	Unit kg	State	Source / reason
1	6	ankle-l, ankle-r, hip-l, hip-r, knee-l...	actuator, datasheet envelope	CubeMars AK80-64 KV80	0.850	sourced	www.cubemars.com/product/ak80-64-kv80-robotic-actuator.h
2	1	armor	armor cage (spec 4.2)	-	unknown	not fitted	Milestone 1 does not fit an armor cage: spec 4.1 mission (stand, walk, carry the payload, sense, stop safely)...
3	1	battery	battery 13S5P (cells only; BMS/enclosure separate)	Molicel INR-21700-P45B	5.105	computed	65 cells x 0.07 kg (molice!-p45b) + enclosure 0.555 kg (aluminum-6061-t6 1.5 mm sheet box 296 x 116 x 83 mm, ...
4	1	bms	battery BMS board (13S, ~60 A class) and cell interconnects (not selected)	envelope only (no material)	0.300	estimated	estimated: engineering estimate: a 13S BMS board for ~60 A (one leg's three actuator peaks, 57 A) with its FE...
5	1	bracket	rect tube 25x25x1.65 mm wall, L 400 mm	aluminum-6061-t6-b221	0.169	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
6	1	camera	RealSense D455	envelope only (no material)	0.116	sourced	www.realsenseai.com/wp-content/uploads/2026/03/RealSense-D4
7	1	compute	Advantech MIC-733-A06A1 (Jetson AGX Orin 64G, fanless): NVIDIA Jetson AGX ...	envelope only (no material)	4.500	sourced	csi.pl/wp-content/uploads/2023/09/MIC-733-AO_D
8	1	fan-1	controller/driver bay fan 1: Sanyo Denki San Ace 80 9GA0824P4H001 (24 V)	envelope only (no material)	0.110	sourced	sy-dep-epc-lpc.web.cern.ch/components/datasheets/epc-lpc%20(conver
9	1	fan-2	controller/driver bay fan 2: Sanyo Denki San Ace 80 9GA0824P4H001 (24 V)	envelope only (no material)	0.110	sourced	sy-dep-epc-lpc.web.cern.ch/components/datasheets/epc-lpc%20(conver
10	1	fasteners	fasteners (actuator, plate, tube-clamp and panel bolts)	-	0.500	estimated	estimated: engineering estimate: ~150 M4-M6 steel screws, nuts and washers (? 2-5 g each) for 6 actuators x 2...
11	2	foot-l, foot-r	plate 360x180x12 mm	aluminum-6061-t6-b221	2.100	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
12	1	imu	Xsens MTI-630 AHRS	envelope only (no material)	0.009	sourced	www.xsens.com/hubfs/MTI%20product%20leaflet_07022025_
13	1	lidar	Velodyne VLP-16 Puck	envelope only (no material)	0.830	sourced	docs.clearpathrobotics.com/assets/files/clearpath_robotics_020972-
14	1	mast	rect tube 25x25x1.65 mm wall, L 210 mm	aluminum-6061-t6-b221	0.089	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
15	1	mount-ankle-la	ankle l actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
16	1	mount-ankle-lb	ankle l actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
17	1	mount-ankle-ra	ankle r actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
18	1	mount-ankle-rb	ankle r actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
19	1	mount-hip-la	hip l actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
20	1	mount-hip-lb	hip l actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
21	1	mount-hip-ra	hip r actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
22	1	mount-hip-rb	hip r actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
23	1	mount-knee-la	knee l actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
24	1	mount-knee-lb	knee l actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
25	1	mount-knee-ra	knee r actuator side plate a	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
26	1	mount-knee-rb	knee r actuator side plate b	aluminum-6061-t6-b221	0.162	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
27	2	panel-rail-l, panel-rail-r	rect tube 25x25x1.65 mm wall, L 1000 mm	aluminum-6061-t6-b221	0.423	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
28	4	panel-standoff-lhi, panel-standoff-llo...	rect tube 25x25x1.65 mm wall, L 131 mm	aluminum-6061-t6-b221	0.055	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
29	1	payload	payload (user requirement)	envelope only (no material)	20.000	requirement	Milestone 1 demo requirement: carry a stated 20 kg payload
30	2	pelvis, shoulder	rect tube 51x51x3.17 mm wall, L 560 mm	aluminum-6061-t6-b221	0.915	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
31	4	shin-l, shin-r, thigh-l, thigh-r	rect tube 51x51x3.17 mm wall, L 780 mm	aluminum-6061-t6-b221	1.274	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
32	1	solar	Renogy RNG-100D-SS (back-mounted, vertical, facing south when parked)	envelope only (no material)	6.400	sourced	store-fhnch.mybigcommerce.com/content/RNG-100D-SS/RNG-100D-SS-G3%20Da
33	2	torso-l, torso-r	rect tube 51x51x3.17 mm wall, L 800 mm	aluminum-6061-t6-b221	1.306	computed	mass.part (aluminum-6061-t6-b221, specified minimum)
34	1	wiring	wiring harness (actuator power + CAN)	-	0.595	estimated	estimated: copper computed: 25.928 m of conductor (2 x AWG 14 per actuator for 19 A peaks, 2 x AWG 24 CAN per...

Known mass 60.68 kg: sourced 17.17, computed 22.11, estimated 1.39, requirement 20.00 kg (mass.budget).

Mass band 60.11-61.89 kg (estimate ranges and sourced tolerances).

Not fitted in this design: armor (see BOM).

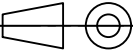
Mass budget closed: every body has a mass with a state.

CG AND SUPPORT-POLYGON CHECK

CG of the known mass (x, y, z)	(67, 0, 1761) mm
Support polygon (contact hull)	4 corners, foot-l + foot-r
Stability margin (known mass)	113 mm
Required margin	>= 50 mm (R-stability)
Unknown-mass items	none
CHECK	PASS

KEY DIMENSIONS (value, basis, where it comes from)

GENERAL ARRANGEMENT - Sentinel M1 ground prototype

Drawing no. CK-GA-SENTINEL-M1 Sheet 2 of 2	Revision
Scale 1:20 (A3) Units mm First-angle projection	R-BD6B4261 
Model: sentinel: 46 part boxes + BOM (34 lines) + mass.budget@sentinel + stability.s...	model hash bd6b42619415a96d716a67d42b3d6062
SCREENING - NOT FOR MANUFACTURE Generated by ConKay drawing.ga-assembly 1.0.0 from the design graph; revision = model hash	ec93e07e11197396d7a5a40c414d1c41